

Machine Learning

Lecture 3: Probabilistic Inference

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We flip the same coin 10 times:



Probability that the next coin flip is **T**?

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~ 0 ~ 0.3 ~ 0.38 ~ 0.5 ~ 0.76 ~ 1

30%?

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$$p_i(F_i = \text{T}) = \theta_i$$

To denote that the probability distribution depends on θ_i , we write

$$p_i(F_i = \text{T} \mid \theta_i) = \text{Ber}(F_i = \text{T} \mid \theta_i) = \theta_i$$

i.e. $F_i \sim \text{Ber}(\theta_i)$

Note the i in the index! We are trying to reason about θ_{11} .

Section 1

Maximum likelihood estimation

All the randomness of a sequence of flips is governed (*modeled*) by the parameters $\theta_1, \dots, \theta_{10}$:

$$p(\textcircled{\text{H}}\textcircled{\text{T}}\textcircled{\text{H}}\textcircled{\text{H}}\textcircled{\text{T}}\textcircled{\text{H}}\textcircled{\text{H}}\textcircled{\text{H}}\textcircled{\text{T}}\textcircled{\text{H}} \mid \theta_1, \theta_2, \dots, \theta_{10})$$

What do we know about $\theta_1, \dots, \theta_{10}$? Can we infer something about θ_{11} ?
At first sight, there is no connection.

Find θ_i 's such that that $p(\textcircled{\text{H}}\textcircled{\text{T}}\textcircled{\text{H}}\textcircled{\text{H}}\textcircled{\text{T}}\textcircled{\text{H}}\textcircled{\text{H}}\textcircled{\text{H}}\textcircled{\text{T}}\textcircled{\text{H}} \mid \theta_1, \theta_2, \dots, \theta_{10})$ is as high as possible. This is a very important principle:

Maximize the *likelihood* of our observation. (*Maximum Likelihood*)

$$???? \quad p(\textcircled{\text{H}}\textcircled{\text{T}}\textcircled{\text{H}}\textcircled{\text{H}}\textcircled{\text{T}}\textcircled{\text{H}}\textcircled{\text{H}}\textcircled{\text{H}}\textcircled{\text{T}}\textcircled{\text{H}} \mid \theta_1, \theta_2, \dots, \theta_{10}) \quad ????$$

We need to model this.

First assumption: The flips are qualitatively the same—identical distribution.

$$p_i(F_i = f_i \mid \theta_i) = p(F_i = f_i \mid \theta)$$

Notice that the parameter θ does not depend on i .

Second assumption: The coin flips do not affect each other—independence.

$$\begin{aligned} & p(\textcircled{\text{H}}\textcircled{\text{T}}\textcircled{\text{H}}\textcircled{\text{H}}\textcircled{\text{T}}\textcircled{\text{H}}\textcircled{\text{H}}\textcircled{\text{H}}\textcircled{\text{T}}\textcircled{\text{H}} \mid \theta) \\ &= p(F_1 = \textcircled{\text{H}} \mid \theta) \cdot p(F_2 = \textcircled{\text{T}} \mid \theta) \cdot \dots \cdot p(F_{10} = \textcircled{\text{H}} \mid \theta) \\ &= \prod_{i=1}^{10} p(F_i = f_i \mid \theta) \end{aligned}$$

In total: The 10 flips are *independent and identically distributed (i.i.d.)*.

Remember θ_{11} ? With the i.i.d. assumption we can link it to $\theta_1, \dots, \theta_{10}$.

Now we can write down the probability of our sequence with respect to θ :

$$\begin{aligned}\prod_{i=1}^{10} p(F_i = f_i \mid \theta) &= (1 - \theta)\theta(1 - \theta)(1 - \theta)\theta(1 - \theta)(1 - \theta)(1 - \theta)\theta(1 - \theta) \\ &= \theta^3(1 - \theta)^7\end{aligned}$$

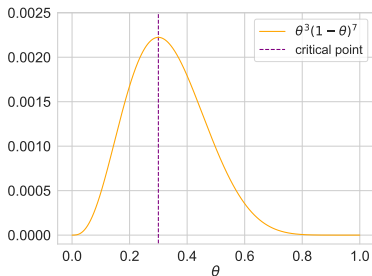
Under our model assumptions (i.i.d.):

$$p(\underbrace{(\text{H T H H T H H H T H})}_{\text{observed data, } \mathcal{D}} \mid \theta) = \theta^3 (1 - \theta)^7$$

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This can be interpreted as a function $f(\theta) := p(\mathcal{D} \mid \theta)$. We want to find the maxima (maximum likelihood) of this function.



Our goal

$$\theta_{\text{MLE}} = \arg \max_{\theta \in [0,1]} f(\theta)$$

Very important: the likelihood function is not a probability distribution over θ since $\int p(\mathcal{D} \mid \theta) d\theta \neq 1$ in general.

How do we maximize the likelihood function?

High-school math! Take the derivative $\frac{df}{d\theta}$, set it to 0, and solve for θ . Check these *critical points* by checking the second derivative.

This is possible, but even for our simple $f(\theta)$ the math is rather ugly.

Can we simplify the problem?

How do we maximize the likelihood function?

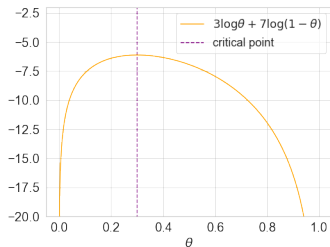
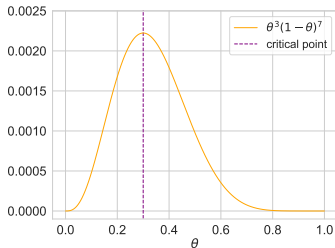
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Can we simplify the problem?

Luckily, monotonic functions preserve critical points.

$$\arg \max_{\theta \in [0,1]} f(\theta) = \arg \max_{\theta \in [0,1]} \log f(\theta)$$



Maximum Likelihood Estimation (MLE) for any coin sequence?

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$$\theta_{\text{MLE}} = \frac{|T|}{|T| + |H|}$$

$|T|, |H|$ denote number of **T**, **H**, respectively.
(derivation in the exercise sheet)

Remember we wanted to find the probability the next coin flip is **T**

$$F_{11} \sim \text{Ber}(\theta_{\text{MLE}})$$

$$p(F_{11} = \text{bT} \mid \theta_{\text{MLE}}) = \text{Ber}(F_{11} = \text{bT} \mid \theta_{\text{MLE}}) = \theta_{\text{MLE}} = \frac{|T|}{|T| + |H|}$$

This justifies 30% as a reasonable answer to our initial question.
Problem solved?!

Just for fun, a totally different sequence (*same coin!*):



$$\theta_{\text{MLE}} = 0.$$

But even a fair coin ($\theta = 0.5$) has 25% chance of showing this result!

The MLE solution seems counter-intuitive. Why?

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We have *prior beliefs*:

"Coins usually don't land heads all the time"

How can we

1. represent such beliefs mathematically?
2. incorporate them into our model?

Section 2

Bayesian inference

How can we represent our beliefs about θ mathematically?

(Subjective) Bayesian interpretation of probability:

Distribution $p(\theta)$ reflects our subjective beliefs about θ .

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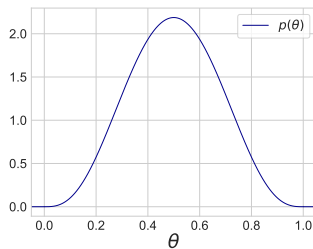
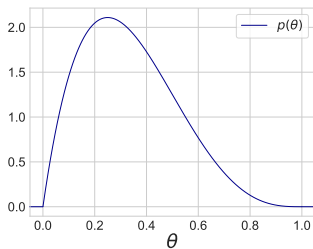
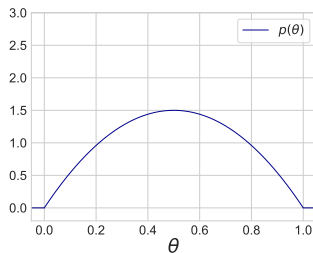
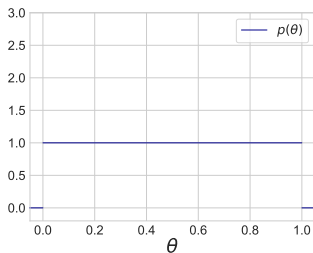
How do we choose $p(\theta)$? The only constraints are

1. It must not depend on the data
2. $p(\theta) \geq 0$ for all θ
3. $\int p(\theta) d\theta = 1$

Properties 2 and 3 have to hold on the support (i.e., feasible values) of θ .
In our setting, only values $\theta \in [0, 1]$ make sense.

This leaves room for (possibly subjective) model choices!

Some possible choices for the prior on θ :



The Bayes formula tells us how to we update our beliefs about θ after observing the data $\mathcal{D}$

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta) \cdot p(\theta)}{p(\mathcal{D})}$$

Here, $p(\theta \mid \mathcal{D})$ is the **posterior** distribution. It encodes our beliefs in the value of θ *after* observing data.

The posterior depends on the following terms:

- $p(\mathcal{D} \mid \theta)$ is the **likelihood**.
- $p(\theta)$ is the **prior** that encodes our beliefs before observing the data.
- $p(\mathcal{D})$ is the **evidence**. It acts as a normalizing constant that ensures that the posterior distribution integrates to 1.

$$\text{posterior} \propto \text{likelihood} \cdot \text{prior}$$

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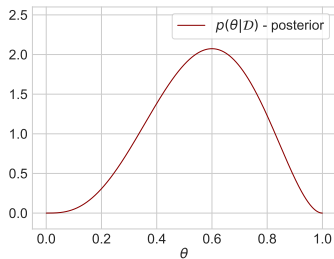
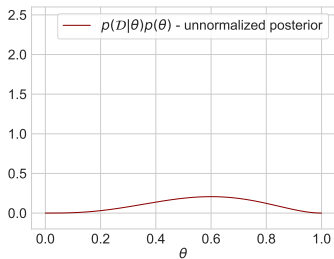
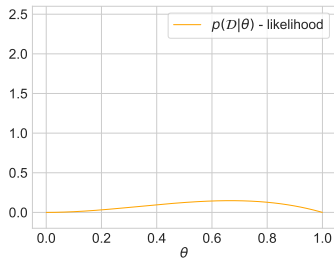
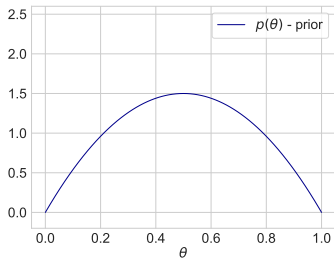
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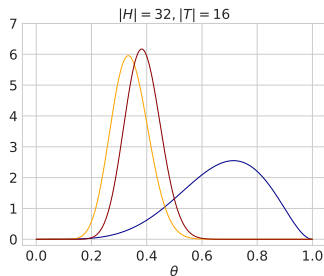
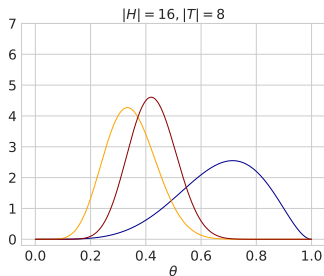
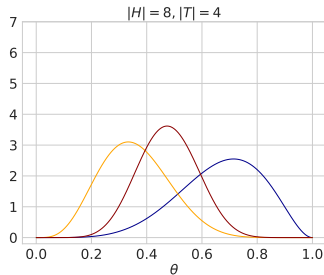
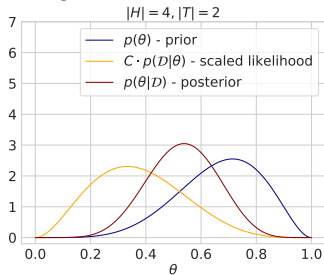
Usually, we define our model by specifying the likelihood and the prior. We can obtain the evidence using the sum rule of probability

$$p(\mathcal{D}) = \int p(\mathcal{D}, \theta) \mathrm{d}\theta = \int p(\mathcal{D} \mid \theta)p(\theta) \mathrm{d}\theta$$

The Bayes formula tells us how to update our beliefs given the data



Observing more data increases our confidence



Note that the likelihood is scaled in the plots for better visibility.

Question: What happens if $p(\theta) = 0$ for some particular θ ?

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Recall:

$$\begin{array}{ccccccc} \text{posterior} & \propto & \text{likelihood} & \cdot & \text{prior} \\ p(\theta \mid \mathcal{D}) & \propto & p(\mathcal{D} \mid \theta) & \cdot & p(\theta) \end{array}$$

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Posterior will always be zero for that particular θ regardless of the likelihood/data.

Section 3

Maximum a posteriori estimation

Back to our coin problem: How do we estimate θ from the data?

In MLE, we were asking the wrong question

$$\theta_{\text{MLE}} = \arg \max_{\theta} p(\mathcal{D} \mid \theta)$$

MLE ignores our prior beliefs and performs poorly if little data is available.

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In MLE, we were asking the wrong question

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MLE ignores our prior beliefs and performs poorly if little data is available.

Actually, we should care about the posterior distribution $p(\theta \mid \mathcal{D})$.

What if we instead maximize the posterior probability?

$$\theta_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid \mathcal{D})$$

This approach is called *maximum a posteriori* (MAP) estimation.

Maximum a posterior estimation

$$\begin{aligned}\theta_{\text{MAP}} &= \arg \max_{\theta} p(\theta \mid \mathcal{D}) \\ &= \arg \max_{\theta} \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})}\end{aligned}$$

We can ignore $\frac{1}{p(\mathcal{D})}$ since it's a (positive) constant independent of θ

$$= \arg \max_{\theta} p(\mathcal{D} \mid \theta)p(\theta)$$

We already know the likelihood $p(\mathcal{D} \mid \theta)$ from before, how do we choose the prior $p(\theta)$?

Often, we choose the prior to make subsequent calculations easier.

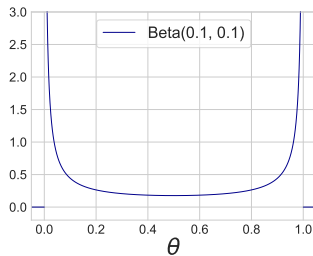
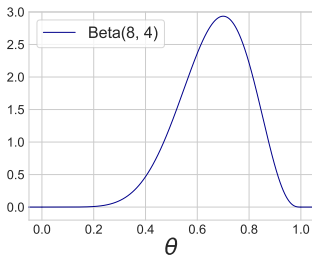
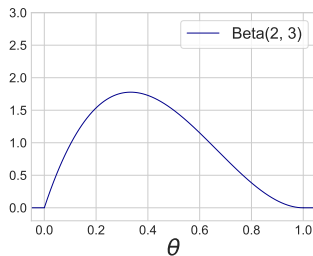
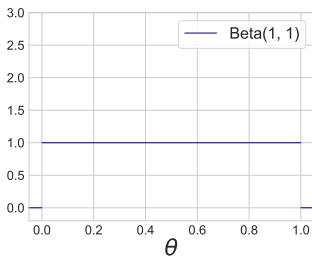
We choose Beta distribution for reasons that will become clear later.

$$\text{Beta}(\theta \mid a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1}, \quad \theta \in [0, 1]$$

where

- $a > 0, b > 0$ are the distribution parameters
- $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$ is the gamma function

The Beta PDF for different choices of a and b :



Let's put everything together

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta) \cdot p(\theta)}{p(\mathcal{D})}$$
$$\propto p(\mathcal{D} \mid \theta) \cdot p(\theta)$$

because $p(\mathcal{D})$ is constant w.r.t. θ .

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because $p(\mathcal{D})$ is constant w.r.t. θ .

We know

$$\begin{aligned} p(\mathcal{D} \mid \theta) &= \theta^{|T|} (1 - \theta)^{|H|}, \\ p(\theta) \equiv p(\theta \mid a, b) &= \frac{\Gamma(a + b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1 - \theta)^{b-1}. \end{aligned}$$

So we get:

$$\begin{aligned} p(\theta \mid \mathcal{D}) &\propto \theta^{|T|} (1 - \theta)^{|H|} \cdot \frac{\Gamma(a + b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1 - \theta)^{b-1} \\ &\propto \theta^{|T|+a-1} (1 - \theta)^{|H|+b-1}. \end{aligned}$$

We are looking for

$$\begin{aligned}\theta_{\text{MAP}} &= \arg \max_{\theta} p(\theta \mid \mathcal{D}) \\ &= \arg \max_{\theta} \theta^{|T|+a-1} (1 - \theta)^{|H|+b-1}\end{aligned}$$

As before, the problem becomes much easier if we consider the logarithm

$$\begin{aligned}\theta_{\text{MAP}} &= \arg \max_{\theta} \log p(\theta \mid \mathcal{D}) \\ &= \arg \max_{\theta} (|T| + a - 1) \log \theta + (|H| + b - 1) \log(1 - \theta)\end{aligned}$$

With some algebra we obtain

$$\theta_{\text{MAP}} = \frac{|T| + a - 1}{|H| + |T| + a + b - 2}$$

Section 4

Estimating the posterior distribution

What we have so far

$$\theta_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid \mathcal{D})$$

The most probable value of θ under the posterior distribution.

Is this the best we can do?

- How certain are we in our estimate?
- What is the probability that θ lies in some interval?

For this, we need to consider the entire posterior distribution $p(\theta \mid \mathcal{D})$, not just its mode θ_{MAP} .

We know the posterior up to a normalizing constant (slide 25)

$$p(\theta \mid \mathcal{D}) \propto \theta^{|T|+a-1} (1 - \theta)^{|H|+b-1}.$$

Finding the true posterior $p(\theta \mid \mathcal{D})$ boils down to finding the normalization constant, such that the distribution integrates to 1.

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Option 1: Brute-force calculation

- Computing $\int_0^1 \theta^{|T|+a-1} (1 - \theta)^{|H|+b-1} d\theta$.
- This is tedious, difficult and boring. Any alternatives?

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Option 2: Pattern matching

- The unnormalized posterior $\theta^{|T|+a-1}(1-\theta)^{|H|+b-1}$ looks similar to the PDF of the Beta distribution

$$\text{Beta}(\theta \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

- Can we use this fact?

The unnormalized posterior

$$p(\theta \mid \mathcal{D}) \propto \theta^{|T|+a-1}(1-\theta)^{|H|+b-1}$$

Beta distribution

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Thus, we can conclude that the appropriate normalizing constant is

$$\frac{\Gamma(|T| + a + |H| + b)}{\Gamma(|T| + a)\Gamma(|H| + b)}$$

and the posterior is a Beta distribution

$$p(\theta \mid \mathcal{D}) = \text{Beta}(\theta \mid a + |T|, b + |H|)$$

Always remember this trick when you try to solve integrals that involve known pdfs (up to a constant factor)!

We started with the following prior distribution

$$p(\theta) = \text{Beta}(\theta \mid a, b)$$

And obtained the following posterior

$$p(\theta \mid \mathcal{D}) = \text{Beta}(\theta \mid a + |T|, b + |H|)$$

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No, this is an instance of a more general principle. Beta distribution is a **conjugate prior** for the Bernoulli likelihood.

If a prior is conjugate for the given likelihood, then the posterior will be of the same family as the prior.

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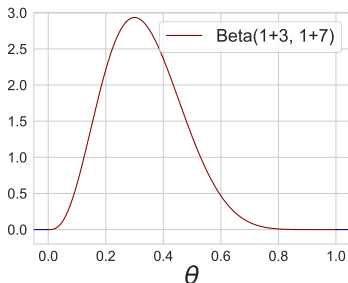
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In our case, we can interpret the parameters a, b of the prior as the number of tails and heads that we saw in the past.

What are the advantages of the fully Bayesian approach?

We have an entire distribution, not just a point estimate



We can answer questions such as:

- What is the expected value of θ under $p(\theta \mid \mathcal{D})$?
- What is the variance of $p(\theta \mid \mathcal{D})$?
- Find a *credible interval* $[\theta_1, \theta_2]$, such that $\Pr(\theta \in [\theta_1, \theta_2] \mid \mathcal{D}) = 95\%$ (not to be confused with frequentist confidence intervals).

We learned about three approaches for parameter estimation:

Maximum likelihood estimation (MLE)

- Task: Solve optimization problem $\max_{\theta} \log p(\mathcal{D} \mid \theta)$
- Result: Point estimate θ_{MLE}
- Coin example: $\theta_{\text{MLE}} = \frac{|T|}{|T|+|H|}$

Maximum a posteriori (MAP) estimation

- Task: Solve optimization problem $\max_{\theta} \log p(\theta \mid \mathcal{D})$
- Result: Point estimate θ_{MAP}
- Coin example: $\theta_{\text{MAP}} = \frac{|T|+a-1}{|T|+|H|+a+b-2}$

Estimating the posterior distribution

- Task: Find the normalizing constant $p(\mathcal{D})$
- Result: Full distribution $p(\theta \mid \mathcal{D})$
- Coin example: $p(\theta \mid \mathcal{D}) = \text{Beta}(\theta \mid a + |T|, b + |H|)$

The three approaches are closely connected.

The posterior distribution is

$$p(\theta \mid \mathcal{D}) = \text{Beta}(\theta \mid a + |T|, b + |H|).$$

Recall that the mode of $\text{Beta}(\alpha, \beta)$ is $\frac{\alpha-1}{\alpha+\beta-2}$, for $\alpha, \beta > 1$.

We see that the MAP solution is the mode of the posterior distribution

$$\theta_{\text{MAP}} = \frac{|T| + a - 1}{|H| + |T| + a + b - 2}$$

If we choose a uniform prior (i.e. $a = b = 1$) we obtain the MLE solution

$$\theta_{\text{MLE}} = \frac{|T| + 1 - 1}{|H| + |T| + 1 + 1 - 2} = \frac{|T|}{|H| + |T|}$$

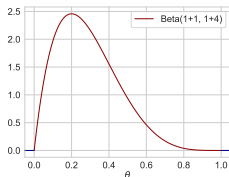
All these nice formulas are a consequence of choosing a conjugate prior. Had we chosen a non-conjugate prior, $p(\theta \mid \mathcal{D})$ and θ_{MAP} could not have a closed form.

How many flips?

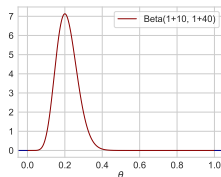
We had $p(\theta \mid \mathcal{D}) = \text{Beta}(\theta \mid a + |T|, b + |H|)$

Visualize the posterior (for given prior, e.g. $a = b = 1$):

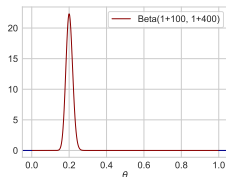
$$|T| = 1, |H| = 4$$



$$|T| = 10, |H| = 40$$



$$|T| = 100, |H| = 400$$



With more data the posterior becomes more peaky – we are more certain about our estimate of θ

Alternative view: a frequentist perspective

For MLE we had $\theta_{\text{MLE}} = \frac{|T|}{|T|+|H|}$

Clearly, we get the same result for $|T| = 1, |H| = 4$ and $|T| = 10, |H| = 40$. Which one is *better*? Why?

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How many flips? Hoeffding's Inequality for a *sampling complexity bound*:

$$p(|\theta_{\text{MLE}} - \theta| \geq \epsilon) \leq 2e^{-2N\epsilon^2} \leq \delta,$$

where $N = |T| + |H|$

For example, I want to know θ , within $\epsilon = 0.1$ error, with probability at least $1 - \delta = 0.99$

We have:

$$N \geq \frac{\ln(2/\delta)}{2\epsilon^2} \rightarrow N \approx 265$$

Section 5

Predicting the next flip

Remember that we want to predict the next coin flip...

For MLE:

1. Estimate $\theta_{\text{MLE}} = \frac{|T|}{|H|+|T|}$ from the data.
2. The probability that the next flip lands tails is

$$p(F_{11} = \text{T} \mid \theta_{\text{MLE}}) = \text{Ber}(F_{11} = \text{T} \mid \theta_{\text{MLE}}) = \theta_{\text{MLE}}$$

Similarly, for MAP:

1. Estimate $\theta_{\text{MAP}} = \frac{|T|+a-1}{|H|+|T|+a+b-2}$ from the data.
2. The probability that the next flip lands tails is

$$p(F_{11} = \text{T} \mid \theta_{\text{MAP}}) = \text{Ber}(F_{11} = \text{T} \mid \theta_{\text{MAP}}) = \theta_{\text{MAP}}$$

What if we have the entire posterior?

We have estimated the posterior distribution $p(\theta \mid \mathcal{D}, a, b)$ of the parameter θ .

Now, we want to compute the probability that the next coin flip is , given observations $\mathcal{D}$ and prior belief a, b :

$$p(F = \text{T} \mid \mathcal{D}, a, b)$$

This distribution is called the **posterior predictive** distribution.

Different from the **posterior over the parameters** $p(\theta \mid \mathcal{D}, a, b)$!

So how do we obtain the posterior predictive distribution?

For simplicity, denote the outcome of the next flip as $f \in \{0, 1\}$.

$$p(F = f \mid \mathcal{D}, a, b) = p(f \mid \mathcal{D}, a, b)$$

We already know the posterior over the parameters $p(\theta \mid \mathcal{D}, a, b)$.

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We already know the posterior over the parameters $p(\theta \mid \mathcal{D}, a, b)$.

Using the sum rule of probability

$$\begin{aligned} p(f \mid \mathcal{D}, a, b) &= \int_0^1 p(f, \theta \mid \mathcal{D}, a, b) d\theta \\ &= \int_0^1 p(f \mid \theta, \mathcal{D}, a, b) p(\theta \mid \mathcal{D}, a, b) d\theta \\ &= \int_0^1 p(f \mid \theta) p(\theta \mid \mathcal{D}, a, b) d\theta \end{aligned}$$

The last equality follows from the conditional independence assumption.

"If we know θ , the next flip f is independent of the previous flips $\mathcal{D}$."

Recall that

$$p(f \mid \theta) = \text{Ber}(f \mid \theta) = \theta^f (1 - \theta)^{1-f}$$

and

$$p(\theta \mid \mathcal{D}, a, b) = \frac{\Gamma(|T| + a + |H| + b)}{\Gamma(|T| + a)\Gamma(|H| + b)} \theta^{|T|+a-1} (1 - \theta)^{|H|+b-1}.$$

Substituting these expressions and doing some (boring) algebra we get

$$\begin{aligned} p(f \mid \mathcal{D}, a, b) &= \int_0^1 p(f \mid \theta) p(\theta \mid \mathcal{D}, a, b) \, d\theta \\ &= \frac{(|T| + a)^f (|H| + b)^{(1-f)}}{|T| + a + |H| + b} \end{aligned}$$

Recall that

$$p(f \mid \theta) = \text{Ber}(f \mid \theta) = \theta^f (1 - \theta)^{1-f}$$

and

$$p(\theta \mid \mathcal{D}, a, b) = \frac{\Gamma(|T| + a + |H| + b)}{\Gamma(|T| + a)\Gamma(|H| + b)} \theta^{|T|+a-1} (1 - \theta)^{|H|+b-1}.$$

Substituting these expressions and doing some (boring) algebra we get

$$\begin{aligned} p(f \mid \mathcal{D}, a, b) &= \int_0^1 p(f \mid \theta) p(\theta \mid \mathcal{D}, a, b) d\theta \\ &= \frac{(|T| + a)^f (|H| + b)^{(1-f)}}{|T| + a + |H| + b} \\ &= \text{Ber} \left(f \mid \frac{|T| + a}{|T| + a + |H| + b} \right) \end{aligned}$$

Note that the posterior predictive distribution doesn't contain θ — we have marginalized it out!

We call this approach fully Bayesian analysis.

Predictions using different approaches

- MLE: $p(F = \textcircled{\text{T}} \mid \theta_{\text{MLE}}) = \text{Ber} \left(F = \textcircled{\text{T}} \mid \frac{|T|}{|T|+|H|} \right)$
- MAP: $p(F = \textcircled{\text{T}} \mid \theta_{\text{MAP}}) = \text{Ber} \left(F = \textcircled{\text{T}} \mid \frac{|T|+a-1}{|T|+a+|H|+b-2} \right)$
- Fully Bayesian: $p(F = \textcircled{\text{T}} \mid \mathcal{D}) = \text{Ber} \left(F = \textcircled{\text{T}} \mid \frac{|T|+a}{|T|+a+|H|+b} \right)$

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Given the prior $a = b = 5$ and the counts $|T| = 4, |H| = 8$

$$p_{\text{MLE}} = \frac{4}{12} \approx 0.33 \quad p_{\text{MAP}} = \frac{8}{20} = 0.40 \quad p_{\text{FB}} = \frac{9}{22} \approx 0.41$$

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How about if we have $|T| = 304, |H| = 306$?

$$p_{\text{MLE}} = \frac{304}{610} \approx 0.50 \quad p_{\text{MAP}} = \frac{308}{618} \approx 0.50 \quad p_{\text{FB}} = \frac{309}{620} \approx 0.50$$

As we observe lots of data, the differences in predictions become less noticeable.

- Maximum likelihood
- Maximum a posteriori
- Fully Bayesian analysis
- Prior, Posterior, Likelihood
- The i.i.d. assumption
- Conjugate prior

- Monotonic transforms for optimization.
- Solving integrals by reverse-engineering densities.

Reading material

- Murphy [ch. 3.1 - 3.3]

Slides are based on an older version by M. Sölch